MIL-P-46964A(MU)
6 DECEMBER 1965
SUPERSEDING
MIL-P-46964(MU)
15 JULY 1963

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, CHEMICAL AGENT GB, M426, WITHOUT FUZE AND BURSTER; FILLING OF

1. SCOPE

1.1 This specification covers the filling of an 8-inch projectile with GB.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-V-13750 — Varnish, Phenol-Formaldehyde, Clear and Aluminum Pigmented.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-1235— Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

PICATINNY ARSENAL

F7548346 — Packing and Marking of Pallet for 8 Inch Projectile, Empty or Loaded.

F8860619 — Projectile, 8 Inch, Gas, Nonpersistent, GB, M426, Filling Assembly. (All drawings referenced in this specification are listed on Dwg F8860619.)

PUBLICATIONS

AMMUNITION PROCUREMENT AND SUPPLY AGENCY

QAI 14-210-1 — Preparation of Ammunition Data Cards.

FSC 1320

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain, prior to their use, that all materials and components have met their respective specifications. The agent shall meet the requirements of the specification under which the agent was manufactured.

3.2 Rust prevention of components. The rust prevention protection on the projectile and components shall not be removed until just prior to filling.

3.3 Filling, closing and marking. The projectile, clean and free from corrosion and foreign matter, shall be filled, closed and marked in accordance with Drawing F8860619.

3.4 Filling level. The liquid level in the filled and closed projectile shall be within the limits in table I when tested as specified in 4.6.1.

TABLE I. *Filling Level*

Temperature range °F		*h	
		Minimum.	Maximum.
Between	40 and 50	7.53	8.22
	50 and 60	7.41	8.09
	60 and 70	7.25	7.97
	70 and 80	7.09	7.87
	80 and 90	6.94	7.75
	90 and 100	6.78	7.63
	100 and 110	6.62	7.47

*h — distance in inches below the nose.
10.0% void at 40°F in minimum projectile.
Expansion factor for GB is 0.00057 ml/°F/ml of agent.

3.5 Closure leakage. The closure of the projectile shall not leak agent when the projectile is laid on its side to assure that the closure is wetted by the agent for a minimum period of 24 hours and tested as specified in 4.6.2.

3.6 Projectile decontamination. The exposed surfaces of the projectile shall be free from GB contamination (see 4.6.3 and 6.4).

3.7 Burster case coating. Any protective coating on the interior of the burster case removed during the decontaminating process shall be replaced. The varnish shall conform to grade A, type I of MIL–V–13750, and shall be a color similar to that on the interior of the burster casing. After the burster casing has been revarnished, the bore shall accept freely a plug gage of 2.650 + 0.002 — 0.000 inch diameter by 25.86 inch minimum length.

3.8 Preproduction. Prior to the start of regular production a preproduction sample of projectiles shall be produced in accordance with this specification for examination and tests (see 4.2.3).

3.9 Workmanship. The projectiles shall be free from visible damage such as dents, cracks or deformation that could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 **Special provisions.**

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting procedure. In case of dispute, the procedures of this specification will govern.

4.2.1.1 *Filling level.* When an alternative procedure is approved for use in determining the filling level, the procedure shall be monitored in accordance with 4.6.1 to insure that the requirements of this specification are satisfied.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Preproduction.* A sample of fifty filled projectiles shall be produced using the same materials, methods, equipment, process and procedure as will be used in production. Any change in materials, methods, process or procedure will require a new preproduction sample. The acceptance number is zero.

4.3 **Lotting.** A lot shall consist of the projectiles filled by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of agent nor more than one weight zone shall be represented in any one lot of filled projectiles.

4.4 **Sampling.**

4.4.1 *Examination.* Sampling shall be conducted in accordance with MIL–STD–105.

4.4.2 *Filling level.* Sampling for filling level shall be in accordance with MIL–STD–1235.

4.5 **Inspection provisions.**

4.5.1 *For examination.* Sample projectiles and the level A preparation for delivery shall be examined in accordance with the classification of defects and MIL–STD–105.

4.5.2 *For filling level.* The inspection for filling level shall be in accordance with 4.6.1 and CSP–1, MIL–STD–1235 using an AQL of 0.25 percent defective.

4.5.3 *For decontaminating process.* Fifty projectiles from the first shift produced under regular production conditions shall be tested as specified in 4.6.3 to qualify the decontaminating procedure or whenever the decontaminating procedure is modified.

4.5.4 *For critical defects.* Each projectile in the lot shall be inspected for critical characteristics listed in the classification of defects.

4.5.5 *Classification of defects.*

4.5.5.1 *Projectile, 8 Inch, Gas Nonpersistent, GB, Filling Assembly (Dwg F8860619).*

Categories	*Defects*	*Acceptance standards*
Critical:		
1	Contamination.	4.6.3
2	Closure leakage.	4.6.2
3	Zone weight marking incorrect by two or more zones.	4.6.4
4	Marking misleading.	

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Categories	*Defects*	*Acceptance standards*
Critical:		
5	Incorrect color code.	
Major:	AQL 1.0 percent defective.	
101	Component missing.	
102	Component incorrectly assembled.	
103	Bourrelet diameter over max after painting.	
104	Threads damaged (see 3.9).	
105	Rotating bands damaged (see 3.9).	
106	Depth from face to top of burster casing incorrect.	
107	Coating missing from adapter threads.	
108	Torque test failure of adapter.	
109	Varnish removed from interior of burster casing.	
110	Lubricant missing from press fit surface (white lead)	
111	Weight zone punch mark inadequate.	
Minor:	AQL 2.5 percent defective.	
201	Marking missing.	
202	Exterior paint missing.	

4.5.5.2 *Preparation for delivery (section 5).*

Categories	*Defects*
Critical:	None defined.
Major:	AQL 2.5 percent defective.
101	Strapping loose or missing.
102	Paliet broken.
103	Marking missing.
104	Grommet missing.
105	Shipping gasket missing.

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4.6 **Tests.**

4.6.1 *Filling level.* The height of the GB in the filled projectile shall be determined radiographically with the projectile oriented in vertical nose up position using Edgewood Arsenal Inspection Standard R13-2.

4.6.2 *Closure leakage.* The filled, sealed and decontaminated projectile shall be laid on its side for the required period of time. The closure of each projectile shall be checked for agent leakage by the use of an approved agent detector.

4.6.3 *Projectile decontamination.* Projectile contamination shall be determined using as approved Edgewood Arsenal GB agent detector system.

4.6.4 *Weight zone.* The projectile shall be weighed on a scale whose accuracy has been certified to one tenth of a pound.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. Unless otherwise specified in the contract or order, the filled and closed projectiles of one zone shall be packed for shipment by palletizing as specified on Drawing F7548346.

5.2 **Data cards.** Preparation of ammunition data cards shall be in accordance with QAI 14-210-1.

5.3 Marking. In addition to any special marking required by the contract or order, palletized shipments shall be marked as specified on above drawing and MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) The Government will furnish radiographic equipment and agent detector system when required.

(c) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility when testing is performed by the Government.

(3) Packing and shipping instructions when applicable.

(4) Time required for the Government to notify the supplier whether or not to proceed with production.

6.2 Filling aid. When filled as specified in 3.4, the projectile carries approximately the weight of GB specified on Drawing F8860619. This information is furnished as an aid to procurement and not as a criterion for filling.

6.3 Caution. Due to the hazards of filling munitions with GB, suppliers are advised to exercise all necessary precautions to protect personnel from injury during filling and handling.

6.4 Decontamination. Decontamination can be accomplished by subjecting the projectile to a spray of decontaminating solution (5 to 10 percent commercial grade soda ash Na_2CO_3) for a minimum of 1 minute and applied at a minimum pressure of 15 pounds per square inch gage (psig) to completely wash all surfaces. The projectile should then be rinsed with water to remove all traces of the solution. The temperature of the spray and rinse water should be maintained at 140° to 160°F. Spray line should have a gage with an accuracy of 1 psig. An approved procedure of control of the solution strength should

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be maintained by periodic examination and the time the projectile is subjected to the spray should be automatically controlled and recorded.

6.5 **Simulant–filled.** Drawing F8860619 is used when projectiles are simulant–filled.

Custodian:

Army—MU

Preparing activity:

Army—MU(EA)

Project No. 1320–A032

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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